GÖDEL–DUMMETT LOGIC OVER A FINITE CATEGORY OF MODELS: THE MODELS OF THE TRANSLATED THEORY ARE STRINGS OF ARROWS, AND A SUBGROUP OBSTRUCTION TO COMMUTING WITH THE IMAGINARIES

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ABSTRACT. Let $\mathcal{X}$ be a finite category in which every endomorphism is invertible, and $\mathcal{P}$ the pretopos of finite set-valued functors on $\mathcal{X}$, a first-order theory whose models are the objects of $\mathcal{X}$. For an intermediate logic L proving the linearity axiom $(p \to q) \vee (q \to p)$ we describe the theory read in first-order logic over L: its models are the strings of non-invertible arrows of $\mathcal{X}$, of length bounded by the depth of L, and its pretopos completion is the category of finite set-valued functors on the category $\mathcal{S}$ of strings, whose arrows are the arrows of $\mathcal{X}$ induced by isomorphisms onto final segments. The translation into L commutes with the imaginaries if and only if every finite functor on $\mathcal{S}$ embeds in one pulled back from $\mathcal{X}$. This holds for classical logic and the logic of the two-element chain. For the other logics it holds for finite posets, the propositional case, for products of a finite poset and a finite group, and when $\mathcal{X}$ has only identity automorphisms; it fails when, for the group H of the automorphisms of the first object of a string which extend to the string and the stabilisers S_1, S_2 of two arrows of $\mathcal{S}$ into it, $\langle H \cap S_1, H \cap S_2 \rangle \neq H \cap \langle S_1, S_2 \rangle$. We give categories with fifteen and seventeen arrows where it fails. The main tool is the linearity of the canonical Kripke model in the presence of elements: two homomorphisms out of a model of the translated theory are comparable.

1. INTRODUCTION

For an intermediate logic L and a small coherent category $\mathcal{C}$ let $\mathsf{H}_L\mathcal{C}$ be the syntactic category of the internal coherent theory of $\mathcal{C}$ in first-order logic over L, and P the pretopos completion, which adds the imaginaries. There is a full and faithful comparison

$$c^L_{\mathcal{C}} \colon \mathsf{H}_L\mathsf{P}\mathcal{C} \longrightarrow \mathsf{P}\mathsf{H}_L\mathcal{C}$$

[6]. For intuitionistic logic it is an equivalence only if $\mathcal{C}$ is Boolean [5]; for the logics proving the linearity axiom it is an equivalence for all propositional

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theories [6], and for the first-order theory of pure equality it is an equivalence only for classical logic [6]. This paper studies the logics L proving linearity over the first-order theories which are closest to the propositional ones: those with finitely many models up to isomorphism, all sorts being finite in each of them.

Let $\mathcal{X}$ be a finite category and $\mathcal{P} = \mathbf{FinSet}^{\mathcal{X}}$ the pretopos of functors from $\mathcal{X}$ to finite sets; if idempotents split in $\mathcal{X}$, the models of $\mathcal{P}$ are the objects of $\mathcal{X}$ and the homomorphisms are the arrows. For a finite poset $\mathcal{X}$ this is a finite propositional theory. We assume that $\mathcal{X}$ is an *EI-category*: every endomorphism is an isomorphism. A *string* is a sequence $s = (m_0 \xrightarrow{a_1} m_1 \to \ldots \xrightarrow{a_r} m_r)$ of non-invertible arrows, $r \geq 0$, and $\mathcal{S}_n$ is the category of the strings with at most n objects, one in each isomorphism class, an arrow $s \to s'$ being an arrow $m_0 \to m_0'$ of $\mathcal{X}$ of the form $\iota_0 a_j \cdots a_1$ for an isomorphism of strings ι from the final segment $(m_j \to \cdots \to m_r)$ onto s'. Let $\pi\colon \mathcal{S}_n \to \mathcal{X}$ be the first object.

Theorem 1.1. *Let $\mathcal{X}$ be a finite EI-category, $\mathcal{P} = \mathbf{FinSet}^{\mathcal{X}}$, and L an intermediate logic proving the linearity axiom, with $n = n_L \in \{1, 2, \ldots, \infty\}$ the supremum of the lengths of the finite chains which are frames of L. Then*

$$\mathsf{PH}_L\mathcal{P} \;\simeq\; \mathbf{FinSet}^{\mathcal{S}_n},$$

the unit of H_L followed by that of P corresponding to $\pi^*\colon \mathbf{FinSet}^{\mathcal{X}} \to \mathbf{FinSet}^{\mathcal{S}_n}$, and $\mathsf{H}_L\mathcal{P}$ to the full subcategory of the subfunctors of the functors π^*A. Consequently $c^L_{\mathcal{P}}$ is an equivalence if and only if every functor $W\colon \mathcal{S}_n \to \mathbf{FinSet}$ is a subfunctor of π^*Y for some $Y\colon \mathcal{X} \to \mathbf{FinSet}$, if and only if the unit $W \to \pi^*\mathrm{Lan}_\pi W$ of the left Kan extension is injective for every W.

For $n = 1$, classical logic, $\mathcal{S}_1$ is a skeleton of the groupoid of the isomorphisms of $\mathcal{X}$. The condition of the theorem is combinatorial. We prove:

- it holds if for all arrows $v\colon t' \to t$, $u\colon t' \to s$ of $\mathcal{S}_n$ every arrow g of $\mathcal{X}$ with $gv = u$ is an arrow $t \to s$ of $\mathcal{S}_n$ (Proposition 6.1); this is the case for finite posets, which gives again the theorem of [6] for finite propositional theories, for the product of a finite poset and a finite group (Corollary 6.2), and for all finite EI-categories if $n_L \leq 2$ (Corollary 6.3);
- more generally it holds if, for every string s, the arrows of $\mathcal{S}_n$ into s are a natural retract of the arrows of $\mathcal{X}$ which are connected to them (Proposition 6.4); this is the case if $\mathcal{X}$ has no automorphisms other than identities (Corollary 6.5);
- it fails if for a string t, with $H \leq G = \mathrm{Aut}_{\mathcal{X}}(m_0)$ the group of the first components of its automorphisms, and two arrows of $\mathcal{S}_n$ into t with stabilisers $S_1, S_2 \leq G$ one has $\langle H \cap S_1, H \cap S_2\rangle \neq H \cap \langle S_1, S_2\rangle$ (Proposition 6.7); this happens in a category with four objects and fifteen arrows (Example 6.8), for every L with $n_L \geq 3$;

- it fails if an arrow g of $\mathcal{X}$ which is not an arrow of $\mathcal{S}_n$ becomes one after composition with two arrows v_1, v_2 of $\mathcal{S}_n$, in two ways which are not connected inside $\mathcal{S}_n$ (Proposition 6.9); a category with seventeen arrows shows that this is not a consequence of the subgroup obstruction (Example 6.10).

For classical logic the assumption on the endomorphisms is not needed: the comparison is an equivalence for every finite category in which idempotents split (Proposition 5.3). The results are summarised as follows, $\mathcal{X}$ being a finite category in which idempotents split.

logic L	$\mathcal{X}$	$c^L_{\mathcal{P}}$ is an equivalence
classical ($n_L = 1$)	any	always
linear, $n_L = 2$	EI	always
linear, $3 \leq n_L \leq \infty$	EI	iff the units $W \to \pi^*\mathrm{Lan}_\pi W$ are injective; not always
intuitionistic	any	only for groupoids [5]

So, in contrast with the propositional case, the Gödel–Dummett translation does not commute with the imaginaries of all these theories, the obstructions are of a group-theoretic nature, and they appear exactly from the three-element chain on. For intuitionistic logic the comparison is an equivalence only if $\mathcal{X}$ is a groupoid [5, Thm. 1.1]; the product of the poset $0 < 1$ and the group $\mathbb{Z}/2$, for which an object outside the image is given in [5, Ex. 7.16], is among the categories for which the comparison is an equivalence for all L proving linearity.

What is new and what is known. New, to the author's knowledge: Theorem 1.1; the linearity of the canonical model with elements (Lemma 3.1) and the description of the models by strings (Proposition 4.2); the classical case (Proposition 5.3); Propositions 6.1, 6.4, 6.7 and 6.9 with their corollaries, and Examples 6.6, 6.8 and 6.10. Known and used: the translation H_L, the comparison and the frames of the logics proving linearity [6]; the completeness theorem of coherent logic and the pretopos completion [4]; Joyal's completeness theorem for Kripke models in the form of a truth lemma for the category of coherent models, proved by the method of diagrams as in [5, Lemma 7.13]; and the fact that the prime filters above a prime filter of a linear Heyting algebra form a chain [3], of which Lemma 3.1 is the version with elements. The description of the models by chains in the propositional case, where $\mathcal{X}$ is a finite poset, is known [1, 2, 6]. EI-categories are standard in the representation theory of finite categories; we use only the definition.

2. The setting

Conventions are those of [6, 5]. Let $\mathcal{X}$ be a finite category in which idempotents split, $\mathcal{P} = \mathbf{FinSet}^{\mathcal{X}}$, $y(x) = \mathcal{X}(x, -)$, and $y(\gamma)\colon y(x') \to y(x)$ the composition with $\gamma\colon x \to x'$. Every A in $\mathcal{P}$ is covered by the morphisms $q_a\colon y(x) \to A$, $a \in A(x)$, $q_a(\varphi) = A(\varphi)(a)$. The models of $\mathcal{P}$ are, up to

isomorphism, the evaluations at the objects of $\mathcal{X}$, and the homomorphisms between these are the arrows of $\mathcal{X}$ [5, §7].

Let L be an intermediate logic and $\mathsf{H}_L\mathcal{P}$ the syntactic category of the internal theory of $\mathcal{P}$ in first-order logic over L [6, §2]: intuitionistic first-order logic with the additional axioms $\lambda(\varphi_1,\dots,\varphi_k)$, for $\lambda(p_1,\dots,p_k)\in L$ and first-order formulas φ_i in a common context, so that the formulas of L are valid in every lattice of subobjects of $\mathsf{H}_L\mathcal{P}$. Its objects are the subobjects $U\le \eta A$, A in $\mathcal{P}$, and we omit η. The subobjects of the objects A of $\mathcal{P}$ in $\mathsf{H}_L\mathcal{P}$ are obtained from those in $\mathcal{P}$ by finite meets and joins, $\Rightarrow$, and f^*, $\exists_f$, $\forall_f$ for the morphisms f of $\mathcal{P}$. Since $\mathcal{P}$ is a pretopos, $\mathsf{H}_L\mathsf{P}\mathcal{P}\simeq\mathsf{H}_L\mathcal{P}$, and $c^L_{\mathcal{P}}$ is the unit $\mathsf{H}_L\mathcal{P}\to\mathsf{P}\mathsf{H}_L\mathcal{P}$; it is an equivalence if and only if $\mathsf{H}_L\mathcal{P}$ is a pretopos [6, Lemma 2.2].

A *model* of $\mathsf{H}_L\mathcal{P}$ is a coherent functor $\hat{G}\colon \mathsf{H}_L\mathcal{P}\to\mathbf{Set}$; it is not required to be a Heyting functor. Its restriction to $\mathcal{P}$ is isomorphic to the evaluation at an object m; we fix such an isomorphism and regard $\hat{G}(U)$, for $U\le A$, as a subset of $A(m)$, and say that $\hat{G}$ is *over* m. A natural transformation $\hat{G}\to\hat{G}'$ of models over m, m' restricts to an arrow $\varphi\colon m\to m'$ and is determined by it; so a *homomorphism* $\hat{G}\to\hat{G}'$ is an arrow $\varphi\colon m\to m'$ of $\mathcal{X}$ with $A(\varphi)(\hat{G}(U))\subseteq\hat{G}'(U)$ for all $U\le A$. Let $\mathcal{K}$ be the category of the models and homomorphisms, with one model in each isomorphism class; the functor $\mathcal{K}\to\mathcal{X}$ is faithful.

Lemma 2.1. *An arrow $\varphi\colon m\to m'$ is a homomorphism $\hat{G}\to\hat{G}'$ if and only if $\varphi\in\hat{G}'(U)$ for every $U\le y(m)$ with $\mathrm{id}_m\in\hat{G}(U)$.*

Proof. If $b\in\hat{G}(U)$, $U\le A$, then $b=q_b(\mathrm{id}_m)$, so $\mathrm{id}_m\in\hat{G}(q_b^*U)$, and $A(\varphi)(b)=q_b(\varphi)$; so $A(\varphi)(b)\in\hat{G}'(U)$ iff $\varphi\in\hat{G}'(q_b^*U)$. □

Lemma 2.2 (truth lemma)**.** *Let $\hat{G}$ be a model over m, $f\colon B\to A$ a morphism of $\mathcal{P}$, $U,V\le B$ in $\mathsf{H}_L\mathcal{P}$, $\alpha\in A(m)$ and $\beta\in B(m)$.*

(a) *$\alpha\in\hat{G}(\forall_f U)$ if and only if for every homomorphism $\varphi\colon\hat{G}\to\hat{G}'$ and every $\beta'\in B(m')$ with $f(\beta')=A(\varphi)(\alpha)$ one has $\beta'\in\hat{G}'(U)$.*

(b) *$\beta\in\hat{G}(U\Rightarrow V)$ if and only if for every homomorphism $\varphi\colon\hat{G}\to\hat{G}'$, $B(\varphi)(\beta)\in\hat{G}'(U)$ implies $B(\varphi)(\beta)\in\hat{G}'(V)$.*

Proof. The implications from left to right hold because φ preserves $\forall_f U$ and $U\Rightarrow V$, and $f^*\forall_f U\le U$, $(U\Rightarrow V)\wedge U\le V$. For the converse of (a) let $\alpha\notin\hat{G}(\forall_f U)$, and consider the classical theory of the coherent functors on $\mathsf{H}_L\mathcal{P}$ with a constant c_b for every element b of the sets $\hat{G}(X)$, X an object of $\mathsf{H}_L\mathcal{P}$, the axioms $g(c_b)=c_{g(b)}$ for the morphisms g of $\mathsf{H}_L\mathcal{P}$, a new constant d of sort B and the axioms $f(d)=c_\alpha$ and $\neg U(d)$. If it had no model, a finite part, written as a subobject θ of a product of A with finitely many objects, with a point in $\hat{G}(\theta)$ over α, would give $V_0:=\exists_{\pi_0}\theta\le A$ with $\alpha\in\hat{G}(V_0)$ and $f^*V_0\le U$ by the completeness theorem for the coherent category $\mathsf{H}_L\mathcal{P}$; then $V_0\le\forall_f U$, contradicting $\alpha\notin\hat{G}(\forall_f U)$. A model of the

theory is a coherent functor $\hat{G}'$ with a natural transformation $\hat{G} \to \hat{G}'$ and an element $\beta' \in B(m') \setminus \hat{G}'(U)$ over the image of α. This is the argument of [5, Lemma 7.13]. (b) is the same, with the axioms $U(c_\beta)$ and $\neg V(c_\beta)$. □

So the evaluation $U \mapsto (\hat{G} \mapsto \hat{G}(U))$ is a Heyting functor $\mathsf{H}_L\mathcal{P} \to \mathbf{Set}^{\mathcal{K}}$, which is conservative by the completeness theorem of coherent logic: if $\hat{G}(U) \subseteq \hat{G}(V)$ for all models, then $U \leq V$.

Kripke models. Let K be a finite poset such that every $\uparrow k$ is a frame of L, and $M\colon K \to \mathcal{X}$ a functor. Then $\mathbf{Set}^K$ is an L-category, and M extends to a Heyting functor $\mathsf{H}_L\mathcal{P} \to \mathbf{Set}^K$ [6, Lemma 2.3]; its value at $U \leq A$ is the family of the subsets of $A(M_k)$ defined by the forcing relation of M. The evaluation at a node k is a model of $\mathsf{H}_L\mathcal{P}$ over M_k, and the transitions $M(k \leq l)$ are homomorphisms between these models.

If L proves the linearity axiom, the finite frames of L are the finite posets in which every $\uparrow k$ is a chain with at most n_L elements [6, Lemma 3.1].

3. Two homomorphisms out of a model are comparable

Lemma 3.1 (linearity)**.** *Let $\mathcal{X}$ be a finite category in which idempotents split, L an intermediate logic proving $(p \to q) \vee (q \to p)$, and $\varphi\colon \hat{G} \to \hat{G}'$, $\psi\colon \hat{G} \to \hat{G}''$ homomorphisms of models of $\mathsf{H}_L\mathcal{P}$. Then there is a homomorphism $\chi\colon \hat{G}' \to \hat{G}''$ with $\chi\varphi = \psi$, or a homomorphism $\chi'\colon \hat{G}'' \to \hat{G}'$ with $\chi'\psi = \varphi$.*

Proof. Let the models be over m, m', m'', and suppose that there is no such χ. The set of the arrows $\chi\colon m' \to m''$ of $\mathcal{X}$ with $\chi\varphi = \psi$ is finite, and none of them is a homomorphism; by Lemma 2.1 there is for each of them $U_\chi \leq y(m')$ with $\mathrm{id}_{m'} \in \hat{G}'(U_\chi)$ and $\chi \notin \hat{G}''(U_\chi)$. Let U' be the meet of the U_χ (the top element, if there is no such arrow), and $E' := \exists_{y(\varphi)}U' \leq y(m)$. Then $\varphi \in \hat{G}'(E')$, since $\varphi = y(\varphi)(\mathrm{id}_{m'})$ and $\mathrm{id}_{m'} \in \hat{G}'(U')$; and $\psi \notin \hat{G}''(E')$, since an element of $\hat{G}''(U')$ over ψ is an arrow χ with $\chi\varphi = \psi$ and $\chi \in \hat{G}''(U')$. If there is no χ' either, we get in the same way $E'' \leq y(m)$ with $\psi \in \hat{G}''(E'')$ and $\varphi \notin \hat{G}'(E'')$. In $\mathsf{H}_L\mathcal{P}$ the subobject $(E' \Rightarrow E'') \vee (E'' \Rightarrow E')$ of $y(m)$ is the top element, so id_m lies in $\hat{G}(E' \Rightarrow E'')$ or in $\hat{G}(E'' \Rightarrow E')$. In the first case $\varphi \in \hat{G}'(E' \Rightarrow E'')$, the homomorphism φ sending id_m to φ; with $\varphi \in \hat{G}'(E')$ this gives $\varphi \in \hat{G}'(E'')$, a contradiction. The second case is symmetric. □

By Lemma 2.2 the forcing at a model $\hat{G}$, in the Kripke model $\mathcal{K}$, involves the homomorphisms out of $\hat{G}$; Lemma 3.1 says that these are totally preordered by $\varphi \preceq \psi$ iff $\psi = \chi\varphi$ for a homomorphism χ.

4. EI-categories: the models are strings

From now on $\mathcal{X}$ is a finite EI-category and L proves linearity. We use the following consequences of the definition. A split monomorphism or epimorphism of $\mathcal{X}$ is invertible: if $\rho\gamma = \mathrm{id}$, then $\gamma\rho$ is an idempotent endomorphism, hence invertible, hence the identity. So idempotents split, and a composite $\delta\gamma$ is invertible only if γ and δ are. If there is a non-invertible arrow $x \to x'$, then x and x' are not isomorphic; so the isomorphism classes of objects are strictly partially ordered by the existence of a non-invertible arrow. The non-invertible arrows out of x form a subfunctor $N_x \leq y(x)$.

Lemma 4.1. *A homomorphism of models of $\mathsf{H}_L\mathcal{P}$ whose underlying arrow is invertible in $\mathcal{X}$ is an isomorphism of models.*

Proof. We show, for all homomorphisms $\chi\colon \hat{G}' \to \hat{G}''$ with invertible underlying arrow at once and all $U \leq A$, that $b \in \hat{G}'(U)$ if $A(\chi)(b) \in \hat{G}''(U)$, by induction on the construction of U. This is clear for the subobjects in $\mathcal{P}$ and compatible with finite meets and joins and f^*; for $\exists_f$ it holds because χ is bijective on every sort. Let $A(\chi)(b) \in \hat{G}''(\forall_f U)$, $f\colon B \to A$; by Lemma 2.2 we have to show $\beta \in \hat{G}'''(U)$ for a homomorphism $\lambda\colon \hat{G}' \to \hat{G}'''$ and β over $A(\lambda)(b)$. By Lemma 3.1 there is a homomorphism μ with $\mu\chi = \lambda$ or with $\mu\lambda = \chi$. In the first case $A(\lambda)(b) = A(\mu)A(\chi)(b) \in \hat{G}'''(\forall_f U)$ and $\beta \in \hat{G}'''(U)$. In the second case λ is a split monomorphism, hence invertible, and so is $\mu\colon \hat{G}''' \to \hat{G}''$; $B(\mu)(\beta)$ lies over $A(\chi)(b)$, so $B(\mu)(\beta) \in \hat{G}''(U)$, and $\beta \in \hat{G}'''(U)$ by the induction hypothesis for μ and U. The case of $U \Rightarrow V$ is the same. □

A *string* is a sequence $s = (m_0 \xrightarrow{a_1} m_1 \to \ldots \xrightarrow{a_r} m_r)$ of non-invertible arrows of $\mathcal{X}$, $r \geq 0$; its *length* is $r+1$, its *final segments* are the strings $s_{\geq j} = (m_j \to \cdots \to m_r)$, and an isomorphism of strings is a family of isomorphisms of the objects commuting with the arrows. The length of a string is at most the number of isomorphism classes of objects of $\mathcal{X}$. The string s is a Kripke model over the chain $0 < \cdots < r$, with $M_j = m_j$; if $r+1 \leq n_L$ let $\hat{G}_s$ be the model of $\mathsf{H}_L\mathcal{P}$ given by the evaluation at the node 0. The evaluation at the node j is $\hat{G}_{s_{\geq j}}$, and $a_j \cdots a_1\colon \hat{G}_s \to \hat{G}_{s_{\geq j}}$ is a homomorphism.

Proposition 4.2. *Let $\mathcal{X}$ be a finite EI-category, L an intermediate logic proving linearity and $n = n_L$.*

(a) *Every model of $\mathsf{H}_L\mathcal{P}$ is isomorphic to $\hat{G}_s$ for a string s of length at most n.*
(b) *The homomorphisms $\hat{G}_s \to \hat{G}_{s'}$ are the arrows $\iota_0 a_j \cdots a_1$, for $0 \leq j \leq r$ and ι an isomorphism of strings from $s_{\geq j}$ onto s'; the index j is determined by s'. In particular $\hat{G}_s \cong \hat{G}_{s'}$ if and only if s and s' are isomorphic strings.*

Proof. Step 1: the string of a model. Let $\hat{G}$ be a model over m_0. By Lemma 3.1 the homomorphisms out of $\hat{G}$ are totally preordered by $\preceq$. If $\varphi \preceq \psi \preceq \varphi$, say $\psi = \chi\varphi$ and $\varphi = \chi'\psi$ with $\chi\colon \hat{G}' \to \hat{G}''$, then $\chi'\chi$ is an endomorphism of m' and $\chi\chi'$ one of m''; so χ is invertible in $\mathcal{X}$, and an isomorphism of models by Lemma 4.1. If $\varphi \prec \psi = \chi\varphi$, then χ is not an isomorphism of models, hence not invertible in $\mathcal{X}$, and the isomorphism class of the codomain of ψ is strictly above that of φ. So there are finitely many classes; choose representatives $\mathrm{id} = \varphi_0 \prec \varphi_1 \prec \cdots \prec \varphi_r$, $\varphi_j\colon \hat{G} \to \hat{G}_j$, and homomorphisms $a_j\colon \hat{G}_{j-1} \to \hat{G}_j$ with $\varphi_j = a_j\varphi_{j-1}$. The a_j are not invertible, and $s := (m_0 \xrightarrow{a_1} \ldots \xrightarrow{a_r} m_r)$ is a string.

Step 2: every homomorphism λ out of $\hat{G}_j$ is $\chi a_{j'} \cdots a_{j+1}$ for some $j' \geq j$ and an isomorphism of models χ. The homomorphism $\lambda\varphi_j$ out of $\hat{G}$ is equivalent to some $\varphi_{j'}$, and $j' \geq j$ because $\varphi_j \preceq \lambda\varphi_j$; so $\lambda\varphi_j = \iota a_{j'} \cdots a_{j+1}\varphi_j$ with an isomorphism of models ι. By Lemma 3.1 at $\hat{G}_j$, λ and $\iota a_{j'} \cdots a_{j+1}$ differ by a homomorphism between their codomains in one of the two directions; it is an endomorphism in $\mathcal{X}$, hence invertible, hence an isomorphism of models.

Step 3: $\hat{G}_j(U)$ is the value of U at the node j of the Kripke model s, for all j and all $U \leq A$, where the forcing relation of s is defined by the usual clauses, whether or not the chain is a frame of L. By induction on the construction of U: both sides agree on the subobjects in $\mathcal{P}$ and are compatible with the coherent operations; for $\forall_f U$, by Lemma 2.2 and Step 2, $\alpha \in \hat{G}_j(\forall_f U)$ iff for all $j' \geq j$, all isomorphisms of models χ out of $\hat{G}_{j'}$ and all β over $A(\chi a_{j'} \cdots a_{j+1})(\alpha)$ we have β in the value of U at the codomain of χ, iff, applying χ^{-1}, for all $j' \geq j$ and all β over $A(a_{j'} \cdots a_{j+1})(\alpha)$ we have $\beta \in \hat{G}_{j'}(U)$, which by induction is the forcing clause at the node j. The case of $\Rightarrow$ is the same.

Step 4: $r+1 \leq n$. Let $U_i \leq y(m_0)$ be the image of $y(a_i \cdots a_1)$, for $1 \leq i \leq r$, the arrows out of m_0 which factor through $a_i \cdots a_1$. The arrow $a_j \cdots a_1$ lies in U_i iff $j \geq i$: if $a_j \cdots a_1 = \beta a_i \cdots a_1$ with $j < i$, then $\beta a_i \cdots a_{j+1}$ is an endomorphism of m_j, so a_{j+1} would be a split monomorphism. So at the element id_{m_0} the U_i are forced exactly at the nodes $j \geq i$, and every valuation of propositional variables by up-sets of the chain $0 < \cdots < r$ is obtained from the U_i, the top and the bottom. For a formula $\lambda(p_1, \ldots, p_k)$ of L and such a valuation, the subobject $\lambda(U_{i_1}, \ldots, U_{i_k})$ of $y(m_0)$ is the top element in $\mathsf{H}_L\mathcal{P}$, so id_{m_0} lies in its value under $\hat{G}$; by Step 3 this value is computed by forcing in the chain, where at the element id_{m_0} it is the propositional forcing of λ. So λ is valid in the chain, the chain is a frame of L, and $r + 1 \leq n$. Hence $\hat{G}_s$ is defined, and $\hat{G} \cong \hat{G}_s$ by Step 3; this is (a).

Step 5: (b). Let s be a string of length at most n. We show first that every homomorphism $\psi\colon \hat{G}_s \to \hat{G}'$ is $\chi a_i \cdots a_1$ with χ an isomorphism of models. Let i be maximal with $a_i \cdots a_1 \preceq \psi$, and $\psi = \lambda a_i \cdots a_1$ with $\lambda\colon \hat{G}_{s_{\geq i}} \to \hat{G}'$; it suffices to show that λ is invertible in $\mathcal{X}$. If $i = r$, the node r is the

top of the chain, so the forcing there is classical and $\mathrm{id}_{m_r} \in \hat{G}_{s_{\geq r}}(\neg N_{m_r})$; then $\lambda \in \hat{G}'(\neg N_{m_r})$, so $\lambda \notin N_{m_r}(m')$. Let $i < r$ and suppose that λ is not invertible. By Lemma 3.1 and the maximality of i there is a homomorphism μ with $\mu\lambda = a_{i+1}$. At the node i of s the element id_{m_i} forces $N_{m_i} \Rightarrow I$, where $I \leq y(m_i)$ is the image of $y(a_{i+1})$: at the nodes $j > i$ it has become $a_j \cdots a_{i+1}$. So $\lambda \in \hat{G}'(N_{m_i} \Rightarrow I)$, and as $\lambda \in N_{m_i}$, $\lambda = \beta a_{i+1}$ for an arrow β of $\mathcal{X}$. Then $a_{i+1} = \mu\beta a_{i+1}$, the endomorphism $\mu\beta$ is invertible, so β and μ are invertible, μ is an isomorphism of models and $a_{i+1} \preceq \lambda$, contradicting the maximality of i.

It remains to show that the isomorphisms of models $\chi_0 \colon \hat{G}_u \to \hat{G}_t$ are the first components of the isomorphisms of strings $u \to t$. An isomorphism of strings is an isomorphism of Kripke models and preserves forcing. Conversely let χ_0 be an isomorphism of models, $u = (m_0 \xrightarrow{a_1} \cdots)$ and $t = (m'_0 \xrightarrow{b_1} \cdots)$. It induces a bijection between the homomorphisms out of $\hat{G}_u$ and out of $\hat{G}_t$ which preserves $\preceq$; by what was shown, the classes are those of the $a_i \cdots a_1$ and of the $b_i \cdots b_1$, which are pairwise inequivalent since their codomains are pairwise non-isomorphic. So u and t have the same length, and $b_1\chi_0 = \chi_1 a_1$ for an isomorphism of models $\chi_1 \colon \hat{G}_{u_{\geq 1}} \to \hat{G}_{t_{\geq 1}}$; by induction on the length χ_1 extends to an isomorphism of strings. The index j in (b) is determined because the codomains of the $a_j \cdots a_1$ are pairwise non-isomorphic. □

Definition 4.3. $\mathcal{S}_n = \mathcal{S}_n(\mathcal{X})$ is the category whose objects are the strings of length at most n, one in each isomorphism class, and whose arrows $s \to s'$ are the arrows $\iota_0 a_j \cdots a_1 \colon m_0 \to m'_0$ of $\mathcal{X}$ of Proposition 4.2(b); $\pi \colon \mathcal{S}_n \to \mathcal{X}$ is the first object, a faithful functor. We write $\mathrm{Aut}_{\mathcal{S}}(s) \leq \mathrm{Aut}_{\mathcal{X}}(m_0)$ for the group of the first components of the automorphisms of the string s.

By Proposition 4.2 the category $\mathcal{K}$ of models of $\mathsf{H}_L\mathcal{P}$ is $\mathcal{S}_n$, a finite category. For a poset $\mathcal{X}$ the strings are the strict chains and $\mathcal{S}_n$ is the poset of the strict chains with at most n elements, ordered by final segments, of [6, §6]; this description of the propositional case by chains is known, from the free Gödel algebras over finite distributive lattices [1] and over bounded distributive lattices in the subvarieties [2].

5. The pretopos completion of the translated theory

Lemma 5.1. *Let A be in $\mathcal{P}$ and S a subfunctor of $\pi^*A \colon \mathcal{S}_n \to$ **FinSet**. Then there is a unique $V \leq A$ in $\mathsf{H}_L\mathcal{P}$ with $\hat{G}_s(V) = S(s)$ for all s.*

Proof. Uniqueness is the conservativity of the evaluation. For a string s with first object m, a string t and an arrow $\varphi \colon m \to t_0$ which is not an arrow $s \to t$ of $\mathcal{S}_n$ there is, by Lemma 2.1, $U \leq y(m)$ with $\mathrm{id}_m \in \hat{G}_s(U)$ and $\varphi \notin \hat{G}_t(U)$. Let $\tau_s \leq y(m)$ be the meet of these finitely many U. Then $\varphi \in \hat{G}_t(\tau_s)$ if and only if φ is an arrow $s \to t$ of $\mathcal{S}_n$. Let V be the join of the $\exists_{q_a}\tau_s$ over the strings s and the $a \in S(s)$. Then $\hat{G}_t(\exists_{q_a}\tau_s)$ is the set of

the $A(\varphi)(a)$, $\varphi\colon s \to t$ in $\mathcal{S}_n$, which is contained in $S(t)$ and contains a if $t = s$. □

Lemma 5.2. *Let $F\colon \mathcal{H} \to \mathcal{E}$ be a coherent functor from a small coherent category to a pretopos which is conservative and full on subobjects, and such that every object of $\mathcal{E}$ is a quotient of a finite coproduct of objects $F(X)$. Then the coherent functor $\mathsf{P}\mathcal{H} \to \mathcal{E}$ extending F is an equivalence.*

Proof. An object of $\mathsf{P}\mathcal{H}$ is a quotient $(\coprod_i X_i)/R$ with R given by subobjects $R_{ij} \leq X_i \times X_j$ in $\mathcal{H}$, and a morphism to $(\coprod_j X'_j)/R'$ is a family of subobjects of the $X_i \times X'_j$ which is a functional relation compatible with R and R'; the same holds in $\mathcal{E}$ for the quotients of the coproducts of the $F(X_i)$. Since F is full on subobjects, every such family in $\mathcal{E}$ comes from $\mathcal{H}$, and since F is conservative, the conditions on the families, which are inequalities between subobjects built by the coherent operations, hold in $\mathcal{H}$ if they hold in $\mathcal{E}$, and two families are equal if their images are. So the extension is full and faithful, and an equivalence relation on $\coprod_i F(X_i)$ in $\mathcal{E}$ comes from one in $\mathsf{P}\mathcal{H}$; hence the extension is essentially surjective. □

Proof of Theorem 1.1. By Lemma 2.2 and Proposition 4.2 the evaluation $F\colon \mathsf{H}_L\mathcal{P} \to \mathbf{FinSet}^{\mathcal{S}_n}$, $F(U)(s) = \hat{G}_s(U)$, is a conservative Heyting functor with $F(\eta A) = \pi^* A$. It is full on subobjects: a subfunctor of $F(U)$, $U \leq A$, is a subfunctor of $\pi^* A$, hence $F(V)$ for some $V \leq A$ by Lemma 5.1, and $V \leq U$ by conservativity. Every $W\colon \mathcal{S}_n \to \mathbf{FinSet}$ is a quotient of a finite coproduct of representable functors $\mathcal{S}_n(s, -)$, and $\mathcal{S}_n(s, -)$ is a subfunctor of $\pi^* y(m_0)$, π being faithful. By Lemma 5.2, $\mathsf{PH}_L\mathcal{P} \simeq \mathbf{FinSet}^{\mathcal{S}_n}$, and $\mathsf{H}_L\mathcal{P}$ corresponds to the full subcategory of the subfunctors of the $\pi^* A$. So $\mathsf{H}_L\mathcal{P}$ is a pretopos, that is $c^L_{\mathcal{P}}$ is an equivalence, if and only if every W is isomorphic to a subfunctor of some $\pi^* A$. A morphism $W \to \pi^* Y$ factors through the unit $W \to \pi^* \mathrm{Lan}_\pi W$, and $\mathrm{Lan}_\pi W$ has finite values, $\mathcal{S}_n$ being finite; so W embeds in some $\pi^* Y$ if and only if the unit is injective. □

For classical logic the models can be described without the assumption on the endomorphisms.

Proposition 5.3 (classical logic)**.** *Let $\mathcal{X}$ be a finite category in which idempotents split, $\mathcal{X}^\times$ the groupoid of its isomorphisms and L classical logic. Then the models of $\mathsf{H}_L\mathcal{P}$ are the objects of $\mathcal{X}$, the homomorphisms are the isomorphisms of $\mathcal{X}$, $\mathsf{PH}_L\mathcal{P} \simeq \mathbf{FinSet}^{\mathcal{X}^\times}$, and $c^L_{\mathcal{P}}$ is an equivalence.*

Proof. The evaluation at an object m is a coherent functor $\mathcal{P} \to \mathbf{Set}$, and $\mathbf{Set}$ is Boolean; so it extends to a Heyting functor on $\mathsf{H}_L\mathcal{P}$, uniquely up to isomorphism [6, §2]. Since $\mathsf{H}_L\mathcal{P}$ is Boolean, every coherent functor on it is Heyting [5, Lemma 2.1]; so there is exactly one model over each m, and the isomorphisms of $\mathcal{X}$ are isomorphisms of models. Let $\varphi\colon m \to m'$ be a homomorphism of models. It preserves complements, hence reflects the subobjects of $\mathsf{H}_L\mathcal{P}$. For an object x of $\mathcal{X}$ and $k \geq 1$ let $E_k \leq 1$ be the image of the complement, in $y(x)^k$, of the union of the diagonals; it

is true in the model over m iff $\mathcal{X}(x,m)$ has at least k elements. As φ preserves and reflects the E_k and reflects the diagonal of $y(x)$, the map $\mathcal{X}(x,m) \to \mathcal{X}(x,m')$ is an injection between sets of the same cardinality, for every x; so φ is invertible. Hence the category $\mathcal{K}$ of models is a skeleton of $\mathcal{X}^{\times}$. The proofs of Lemma 5.1 and of the equivalence $\mathsf{PH}_L\mathcal{P} \simeq \mathbf{FinSet}^{\mathcal{K}}$ in Theorem 1.1 used only Lemma 2.1, the conservativity of the evaluation, the finiteness of $\mathcal{K}$ and the faithfulness of $\pi\colon \mathcal{K} \to \mathcal{X}$, so they apply. Finally let $W\colon \mathcal{K} \to \mathbf{FinSet}$ and m in $\mathcal{K}$; $\mathrm{Lan}_\pi W(m)$ is the colimit of $W(m')$ over the category of the pairs $(m', g\colon m' \to m)$, whose arrows $(m', g'v) \to (m'', g')$ are the isomorphisms v. In it $g'v$ is invertible iff g' is, so the pairs with g invertible form a union of connected components, with the terminal object (m, id), and the unit $W(m) \to \mathrm{Lan}_\pi W(m)$ is injective. □

6. When the translation commutes with the imaginaries

For a string s with first object m let $\mathcal{J}_s$ be the category of the pairs (t,g), t a string and $g\colon t_0 \to m$ an arrow of $\mathcal{X}$, an arrow $(t,g) \to (t',g')$ being an arrow $v\colon t \to t'$ of $\mathcal{S}_n$ with $g'v = g$. Then $\mathrm{Lan}_\pi W(m)$ is the colimit of $W(t)$ over $\mathcal{J}_s$, and the unit at s is the map from $W(s)$, at the object (s, id). Let $\mathcal{J}_s^0$ be the full subcategory of the (t,g) with g an arrow $t \to s$ of $\mathcal{S}_n$; it has the terminal object (s, id), so the colimit of W over $\mathcal{J}_s^0$ is $W(s)$. An arrow of $\mathcal{J}_s$ into an object of $\mathcal{J}_s^0$ has its domain in $\mathcal{J}_s^0$. Let $\mathcal{J}_s^c \supseteq \mathcal{J}_s^0$ be the connected component of (s, id) in $\mathcal{J}_s$, and $C_s(t)$ the set of the arrows g with (t,g) in $\mathcal{J}_s^c$; so $\mathcal{S}_n(t,s) \subseteq C_s(t) \subseteq \mathcal{X}(t_0, s_0)$, and $gv \in C_s(t')$ iff $g \in C_s(t)$, for $v\colon t' \to t$ in $\mathcal{S}_n$.

Proposition 6.1. *Suppose that for all arrows $v\colon t' \to t$ and $u\colon t' \to s$ of $\mathcal{S}_n$, every arrow $g\colon t_0 \to s_0$ of $\mathcal{X}$ with $gv = u$ is an arrow $t \to s$ of $\mathcal{S}_n$. Then $c_{\mathcal{P}}^L$ is an equivalence.*

Proof. The hypothesis says that an arrow of $\mathcal{J}_s$ out of an object of $\mathcal{J}_s^0$ has its codomain in $\mathcal{J}_s^0$. So $\mathcal{J}_s^0$ is a union of connected components of $\mathcal{J}_s$, the colimit of W over $\mathcal{J}_s$ is the disjoint union of $W(s)$ and the colimit over the other components, and the unit is injective. □

Corollary 6.2. *Let X be a finite poset and Γ a finite group, regarded as a category with one object. For $\mathcal{X} = X \times \Gamma$, in particular for $\mathcal{X} = X$, the comparison $c_{\mathcal{P}}^L$ is an equivalence for every L proving linearity.*

Proof. An arrow of $\mathcal{X}$ is a pair $(x \le x', \gamma)$; it is invertible iff $x = x'$, so $\mathcal{X}$ is an EI-category, and a string is a strict chain $x_0 < \cdots < x_r$ with elements $\gamma_1, \ldots, \gamma_r$. For strings t, s whose chains are $x_0 < \cdots < x_r$ and $x_j < \cdots < x_r$, every $\gamma_0 \in \Gamma$ is the first component of a unique isomorphism of strings from $t_{\ge j}$ onto s, the other components being determined successively. So the arrows $t \to s$ of $\mathcal{S}_n$ are all the arrows $t_0 \to s_0$ of $\mathcal{X}$ if the chain of s is a final segment of that of t, and there are none otherwise. Let $v\colon t' \to t$, $u\colon t' \to s$ and $g\colon t_0 \to s_0$ with $gv = u$. The chains of t and s are final segments of that

of t', and the first element of t is below that of s; so the chain of s is a final segment of that of t, and g is an arrow of $\mathcal{S}_n$. Proposition 6.1 applies. □

For $\Gamma = 1$ Corollary 6.2 is the theorem of [6] that the translation into a logic proving linearity commutes with the imaginaries of the finite propositional theories. For $X = \{0 < 1\}$ and $\Gamma = \mathbb{Z}/2$ the pretopos $\mathcal{P}$ is that of [5, Ex. 7.16], a proposition p and a sort with exactly two elements and decidable equality; over intuitionistic logic a twisted form of this sort is not in the image of the comparison, and it needs a Kripke model over a fork.

Corollary 6.3. *If $n_L \leq 2$, in particular for classical logic and for the logic of the two-element chain, $c^L_{\mathcal{P}}$ is an equivalence for every finite EI-category $\mathcal{X}$.*

Proof. Let $v\colon t' \to t$ and $u\colon t' \to s$ be arrows of $\mathcal{S}_n$ and $gv = u$. If v is invertible, $g = uv^{-1}$ is an arrow of $\mathcal{S}_n$. Otherwise $t' = (m_0 \xrightarrow{a} m_1)$, t has length one and $v = \iota a$ with $\iota\colon m_1 \to t_0$ invertible. If u were invertible, a would be a split monomorphism; so $u = \iota' a$ with $\iota'\colon m_1 \to s_0$ invertible, and s has length one. Then $\iota'^{-1} g \iota$ is an endomorphism of m_1, so g is invertible, and an invertible arrow between strings of length one is an arrow of $\mathcal{S}_n$. Proposition 6.1 applies. □

The hypothesis of Proposition 6.1 says that $C_s(t) = \mathcal{S}_n(t, s)$. It is the simplest case of the following.

Proposition 6.4 (retraction)**.** *Suppose that for every string s of length at most n there are maps $r_t\colon C_s(t) \to \mathcal{S}_n(t, s)$, t in $\mathcal{S}_n$, with $r_t(\kappa) = \kappa$ for $\kappa \in \mathcal{S}_n(t, s)$ and $r_{t'}(gv) = r_t(g)v$ for $g \in C_s(t)$ and $v\colon t' \to t$ in $\mathcal{S}_n$. Then $c^L_{\mathcal{P}}$ is an equivalence.*

Proof. The colimit of W over $\mathcal{J}_s$ is the disjoint union of the colimits over the connected components. On the colimit over $\mathcal{J}_s^c$ let ρ send the class of $w \in W(t)$ at the object (t, g) to $W(r_t(g))(w) \in W(s)$. This is well defined: for $v\colon (t', gv) \to (t, g)$ and $w' \in W(t')$ the elements w' at (t', gv) and $W(v)w'$ at (t, g) are both sent to $W(r_t(g)v)(w')$. Since $r_s(\mathrm{id}) = \mathrm{id}$, ρ is a left inverse of the map from $W(s)$ to the colimit over $\mathcal{J}_s^c$, which is a summand of $\mathrm{Lan}_\pi W(s_0)$; so the unit is injective. □

Corollary 6.5. *If every automorphism of $\mathcal{X}$ is an identity, then $c^L_{\mathcal{P}}$ is an equivalence for every L proving linearity.*

Proof. Two arrows $t \to s$ of $\mathcal{S}_n$ are $\iota_0 p$ and $\iota'_0 p$ for the same composite p of arrows of t and isomorphisms of strings ι, ι' onto s; these differ by an automorphism of the string s, whose first component is the identity by assumption. So $\mathcal{S}_n(t, s)$ has at most one element. We show that it is nonempty if $C_s(t)$ is; then the unique maps $r_t\colon C_s(t) \to \mathcal{S}_n(t, s)$ satisfy the conditions of Proposition 6.4. It suffices to show, for $v\colon t' \to t$ in $\mathcal{S}_n$ and $g\colon t_0 \to s_0$ in $\mathcal{X}$ with $\mathcal{S}_n(t', s) \neq \emptyset$, that $\mathcal{S}_n(t, s) \neq \emptyset$, the converse being clear. Up to isomorphism $t = t'_{\geq j}$ and $s = t'_{\geq k}$, with $t' = (m_0 \to \cdots \to m_r)$.

If $k < j$, the composite $p\colon m_k \to m_j$ of the arrows of t' is not invertible, and gp is an endomorphism of m_k, so p would be a split monomorphism. So $k \geq j$, and s is a final segment of t. □

Example 6.6. Let $\mathcal{X}$ have the objects 0, 1, 2 and, besides the identities, the arrows $a\colon 0 \to 1$, $c, c'\colon 1 \to 2$ and $d = ca = c'a\colon 0 \to 2$:

$$0 \xrightarrow{a} 1 \underset{c'}{\overset{c}{\rightrightarrows}} 2$$

By Corollary 6.5, $c^L_{\mathcal{P}}$ is an equivalence for every L proving linearity. The hypothesis of Proposition 6.1 fails for $n \geq 3$: for $v = a\colon (0 \xrightarrow{a} 1 \xrightarrow{c} 2) \to (1 \xrightarrow{c} 2)$ and $u = d\colon (0 \xrightarrow{a} 1 \xrightarrow{c} 2) \to (2)$ one has $c'v = u$, and c' is not an arrow $(1 \xrightarrow{c} 2) \to (2)$ of $\mathcal{S}_n$. The retraction sends c' to c.

Proposition 6.7 (subgroup obstruction)**.** *Let t be a string of length at most n with first object m, $G = \mathrm{Aut}_{\mathcal{X}}(m)$, $H = \mathrm{Aut}_{\mathcal{S}}(t) \leq G$, and let $v_i\colon t'_i \to t$, $i = 1, 2$, be arrows of $\mathcal{S}_n$ with stabilisers $S_i = \{g \in G \mid gv_i = v_i\}$. If*

$$\langle H \cap S_1,\ H \cap S_2\rangle \ \neq\ H \cap \langle S_1, S_2\rangle,$$

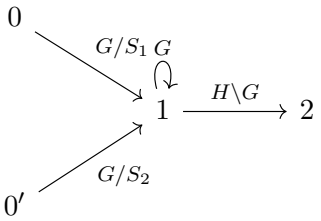
then $c^L_{\mathcal{P}}$ is not an equivalence.

Proof. Let W be the pushout in $\mathbf{FinSet}^{\mathcal{S}_n}$ of the morphisms $\mathcal{S}_n(t, -) \to \mathcal{S}_n(t'_i, -)$, $\kappa \mapsto \kappa v_i$, and $x_i \in W(t'_i)$ the class of the identity; so $W(v_1)x_1 = W(v_2)x_2 =: y$. The arrows $t'_i \to t$ of $\mathcal{S}_n$ are the hv_i, $h \in H$: by Proposition 4.2(b) the index of the final segment is determined, and two isomorphisms of strings onto t differ by an automorphism of t. So $W(t)$ is the quotient of $Hv_1 \sqcup Hv_2$ by the relation generated by $hv_1 \sim hv_2$, and $hv_i = h'v_i$ iff $h^{-1}h' \in S_i$; hence the stabiliser of y in H, for the action of H on $W(t)$ by the arrows $t \to t$, is $\langle H \cap S_1, H \cap S_2\rangle$. Suppose that $\theta\colon W \to \pi^*Y$ is injective, and let $\bar{y} = \theta_t(y) \in Y(m)$. By naturality $\bar{y} = Y(v_i)\theta(x_i)$, so $Y(g)\bar{y} = \bar{y}$ for $g \in S_i$, hence for $g \in \langle S_1, S_2\rangle$. For $h \in H \cap \langle S_1, S_2\rangle$ we get $\theta_t(W(h)y) = Y(h)\bar{y} = \theta_t(y)$, so $W(h)y = y$ and $h \in \langle H \cap S_1, H \cap S_2\rangle$. So under the hypothesis W is not a subfunctor of any π^*Y, and Theorem 1.1 applies. □

Example 6.8. Let $G = \mathbb{Z}/2 \times \mathbb{Z}/2$, $S_1 = \langle(1, 0)\rangle$, $S_2 = \langle(0, 1)\rangle$ and $H = \langle(1, 1)\rangle$. Let $\mathcal{X}$ have the objects 0, 0′, 1, 2, with $\mathrm{Aut}(1) = G$ and trivial automorphism groups otherwise, and the arrows

$$\mathcal{X}(0, 1) = G/S_1, \qquad \mathcal{X}(0', 1) = G/S_2, \qquad \mathcal{X}(1, 2) = H\backslash G,$$
$$\mathcal{X}(0, 2) = \{d\}, \qquad \mathcal{X}(0', 2) = \{d'\},$$

composition being given by the actions of G on the cosets and by the unique maps to the one-element sets:



This is a category with fifteen arrows, and an EI-category. Let $a = S_1 \in \mathcal{X}(0,1)$, $a' = S_2 \in \mathcal{X}(0',1)$ and $c = H \in \mathcal{X}(1,2)$. The string $t = (1 \xrightarrow{c} 2)$ has $\mathrm{Aut}_{\mathcal{S}}(t) = \{g \mid cg = c\} = H$, and the arrows $a\colon (0 \xrightarrow{a} 1 \xrightarrow{c} 2) \to t$ and $a'\colon (0' \xrightarrow{a'} 1 \xrightarrow{c} 2) \to t$ of $\mathcal{S}_3$ have the stabilisers S_1 and S_2. Here $H \cap S_1 = H \cap S_2 = 1$ and $\langle S_1, S_2\rangle = G$; so by Proposition 6.7 the comparison $c^L_{\mathcal{P}}$ is not an equivalence for the logics L proving linearity with $n_L \geq 3$, in particular for the Gödel–Dummett logic. In terms of Kripke models: the functor W of the proof has two points at a node of type t, exchanged by $(1,1)$; a point of this kind which comes from a node below along a and from a node below along a' must, in a coherent imaginary, be fixed by $(1,0)$ and by $(0,1)$.

The same construction gives an obstruction which does not refer to automorphisms of the first object.

Proposition 6.9 (one-step obstruction)**.** *Let s, t be strings of length at most n, $g\colon t_0 \to s_0$ an arrow of $\mathcal{X}$, and $v_i\colon t_i \to t$, $i = 1,2$, arrows of $\mathcal{S}_n$ such that gv_1 and gv_2 are arrows $t_i \to s$ of $\mathcal{S}_n$. Let $\sim$ be the equivalence relation on $\mathcal{S}_n(t_1, s) \sqcup \mathcal{S}_n(t_2, s)$ generated by $\kappa v_1 \sim \kappa v_2$, $\kappa \in \mathcal{S}_n(t, s)$. If $gv_1 \not\sim gv_2$, then $c^L_{\mathcal{P}}$ is not an equivalence.*

Proof. Let W be the pushout of the morphisms $\mathcal{S}_n(t,-) \to \mathcal{S}_n(t_i,-)$, $\kappa \mapsto \kappa v_i$, and $x_i \in W(t_i)$ the classes of the identities, so that $W(v_1)x_1 = W(v_2)x_2 =: y$. Then $W(s)$ is the quotient of $\mathcal{S}_n(t_1, s) \sqcup \mathcal{S}_n(t_2, s)$ by $\sim$, and the elements $z_i := W(gv_i)x_i$, the classes of the gv_i, are different. For a morphism $\theta\colon W \to \pi^*Y$ we have $\theta_s(z_i) = Y(gv_i)\theta_{t_i}(x_i) = Y(g)Y(v_i)\theta_{t_i}(x_i) = Y(g)\theta_t(y)$ for $i = 1,2$; so θ is not injective, and Theorem 1.1 applies. □

If g is an arrow $t \to s$ of $\mathcal{S}_n$, then $gv_1 \sim gv_2$; so the hypothesis implies that (t, g) is an object of $\mathcal{J}^c_s$ outside $\mathcal{J}^0_s$, and that there is no retraction as in Proposition 6.4, which would give $gv_i = r_t(g)v_i$.

Example 6.10. Let $G = \mathbb{Z}/2 \times \mathbb{Z}/2$, $a = (1,0)$, $b = (0,1)$. Let $\mathcal{X}$ have the objects 0, 0′, 1, 2, with $\mathrm{Aut}(2) = \langle b\rangle$ and trivial automorphism groups otherwise, and the arrows

$$\mathcal{X}(0,1) = \mathcal{X}(0,2) = G/\langle a\rangle, \qquad \mathcal{X}(1,2) = G,$$
$$\mathcal{X}(0',1) = \mathcal{X}(0',2) = G/\langle a+b\rangle,$$

composition being the addition of G. This is an EI-category with seventeen arrows. Let $s = (2)$, $t = (1 \xrightarrow{0} 2)$, so that $\mathcal{S}_3(t, s) = \{0, b\}$, and $g = a \in \mathcal{X}(1, 2)$. Let $v_1 = \langle a \rangle \in \mathcal{X}(0, 1)$ and $v_2 = \langle a + b \rangle \in \mathcal{X}(0', 1)$, arrows of $\mathcal{S}_3$ into t from the strings $t_1 = (0 \xrightarrow{v_1} 1 \xrightarrow{0} 2)$ and $t_2 = (0' \xrightarrow{v_2} 1 \xrightarrow{0} 2)$. Then $gv_1 = a + \langle a \rangle = 0 \cdot v_1$ and $gv_2 = a + \langle a + b \rangle = b \cdot v_2$ are arrows $t_i \to s$ of $\mathcal{S}_3$. The relation $\sim$ is generated by $0 \cdot v_1 \sim 0 \cdot v_2$ and $b \cdot v_1 \sim b \cdot v_2$, and $b \cdot v_i \neq 0 \cdot v_i$; so $gv_1 \not\sim gv_2$, and by Proposition 6.9 the comparison is not an equivalence for $n_L \geq 3$. Proposition 6.7 does not apply to this category: the only object with a non-trivial automorphism is 2, the only string with first object 2 is (2), and for it $H = \mathrm{Aut}_{\mathcal{X}}(2)$, so that $H \cap S_i = S_i$.

7. Questions

Question 7.1. Give a closed description of the finite EI-categories $\mathcal{X}$ for which every finite functor on $\mathcal{S}_n(\mathcal{X})$ embeds in one pulled back from $\mathcal{X}$. Is the existence of the retractions of Proposition 6.4 necessary? Is the absence of the obstruction of Proposition 6.9, for all finite families of arrows v_i into a string, sufficient?

Question 7.2. Describe the models of $\mathsf{H}_L\mathcal{P}$ when $\mathcal{X}$ is a finite category with split idempotents which is not an EI-category. Lemma 3.1 still holds, but a non-invertible endomorphism has a power which is a non-identity idempotent, whose splitting is a pair of non-invertible arrows with identity composite, so that there are strings of unbounded length. A related example outside the present setting, its models being infinite: for the theory of pure equality the comparison is an equivalence only for classical logic [6].

Question 7.3. For the logics which prove the weak excluded middle and not linearity the comparison fails already for the four-element chain [6]. Is there a description of the models of the translated theory over a finite poset, in which the chains are replaced by the finite rooted frames of the logic?

Declarations

The author has no competing interests to declare.

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